

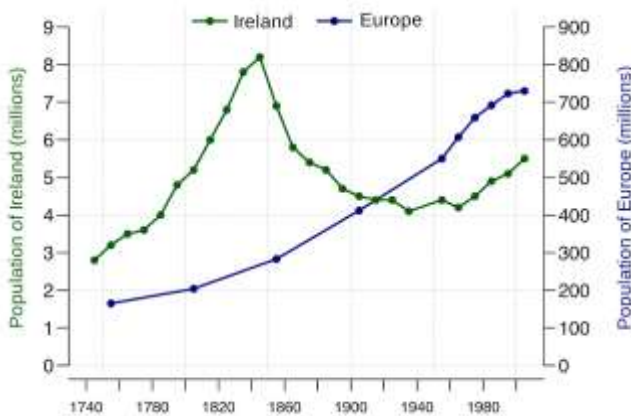


JAL2521 TIME SERIES ANALYSIS AND FORECASTING

UNIT II **STATISTICS BACKGROUND FOR FORECASTING**

UNIT II **STATISTICS BACKGROUND FOR FORECASTING** **SYLLABUS**

Graphical Displays - Time Series Plots - Plotting Smoothed Data - Numerical Description of Time Series Data - Use of Data Transformations and Adjustments - General Approach to Time Series Modelling and Forecasting - Evaluating and Monitoring Forecasting Model Performance.



GRAPHICAL DISPLAYS

- Time Series Plots
- Plotting Smoothed Data

INTRODUCTION

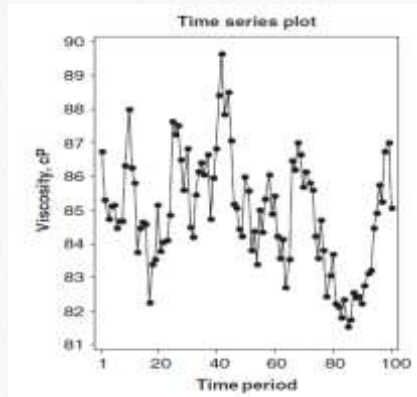
- Forecasts are based on data or observations on the variable of interest → uses data in the form of a time series.
- Observation on a variable at time period t is denoted by y_t , $t = 1, 2, \dots, T$, where $T \rightarrow$ the most recent time period.
- The variable → eg. temperature, rainfall, pollution rate, product sales, product demand, stock prices, unemployment rate, electricity consumption, patient heart rate, app downloads, traffic, water levels, forest coverage (deforestation)
- Variable can represent an instantaneous quantity (10, 8, 20, 8, 11) or a cumulative quantity (10, 18, 38, 46, 57).
- 2 types of values:
 1. Forecast or predicted value of $y_t \rightarrow$ value made at some previous time period, say, $t - \tau$, where $\tau \rightarrow$ forecast lead time $\rightarrow$ predictions for the Jan to Dec 2026 sales.
 2. Fitted value of $y_t \rightarrow$ value that has resulted from estimating the parameters in a time series model to historical data – eg. predictions for the Jan to Dec 2024 sales compared with actual sales in those months.

INTRODUCTION

- Observation made at time period $t \rightarrow y_t$
- Forecast made at time period $t - \tau \rightarrow \hat{y}_t(t - \tau)$.
- Forecast made at time period $t - 1 \rightarrow \hat{y}_t(t - 1) \rightarrow$ Lead-1 forecast
- Fitted value of $y_t \rightarrow \hat{y}_t$.
- Forecast errors $\rightarrow$ error that results from a forecast of y_t that was made at time period $t - \tau \rightarrow$ lead $-\tau$ forecast error.
- Forecast error = observed value - forecast value $e_t(\tau) = y_t - \hat{y}_t(t - \tau)$
- Eg. the difference between observed value of Dec 2024 and forecast value of Dec 2024 sales made earlier (eg. Nov 2024.)
- Eg. if $\tau=1$, lead-1 error is $e_t(1) = y_t - \hat{y}_t(t - 1)$
- Residual $\rightarrow$ The difference between the observation y_t and the value obtained by fitting a time series model to the data, or a fitted value $\hat{y}_t$ defined earlier.
- Residual = observed value - fitted value. $e_t = y_t - \hat{y}_t$
- In practice, residual < forecast errors

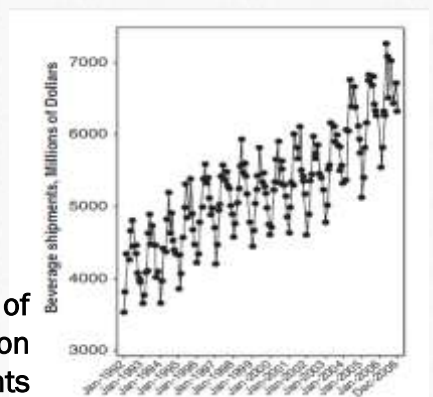
GRAPHICAL DISPLAYS - Time Series Plots

- Developing a forecasting model should always begin with graphical display and analysis of the available data.
- Basic graphical display for time series data is the time series plot $\rightarrow$ graph of y_t versus the time period, t , for $t = 1, 2, \dots, T$.
- Easy to visualize trend and seasonality.



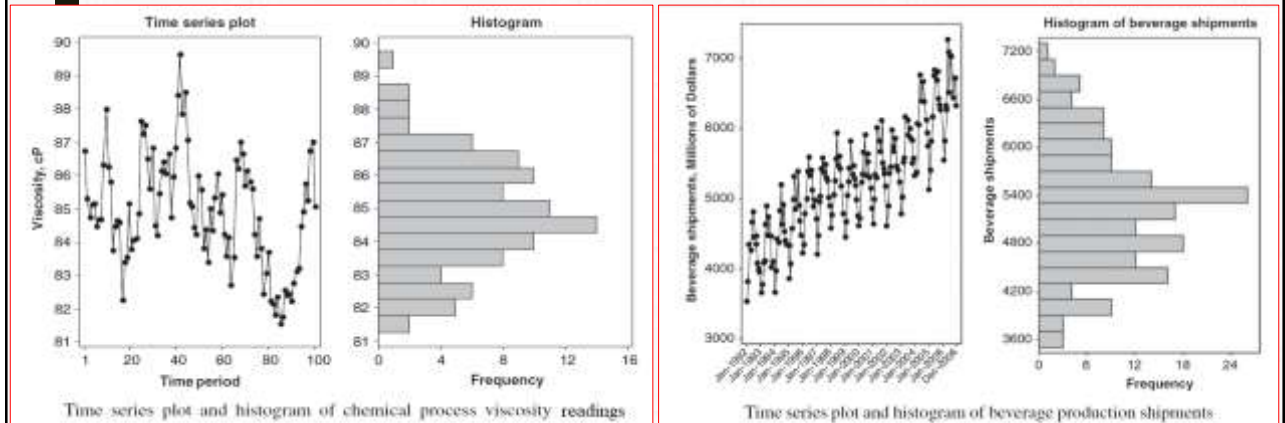
Time series plot of chemical process viscosity readings

Time series plot of beverage production shipments



GRAPHICAL DISPLAYS - Time Series Plots

- Histogram summarizes the data across the time dimension where the key time-dependent features of the data are lost, hence not so useful. Similarly stem-and leaf plots and boxplots are also not so useful.



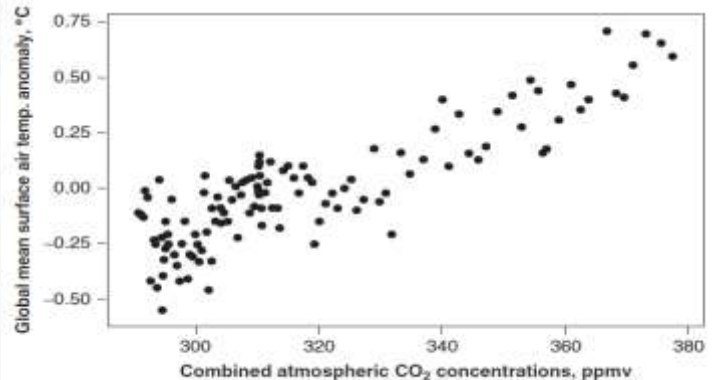
GRAPHICAL DISPLAYS - Time Series Plots

- Histogram summarizes the data across the time dimension where the key time-dependent features of the data are lost, hence not so useful. Similarly stem-and leaf plots and boxplots are also not so useful.

Stem	Leaf
4	4 6 7 9
5	
6	3 4 6 8 8
7	2 2 5 6
8	1 4 8
9	
10	6

GRAPHICAL DISPLAYS - Time Series Plots

- In case of multiple variables of interest, scatter plots can be useful in displaying the relationship between the variables.
- Scatter plot cannot establish a causal relationship between two variables- useful in displaying how the variables have varied together in the historical data set.

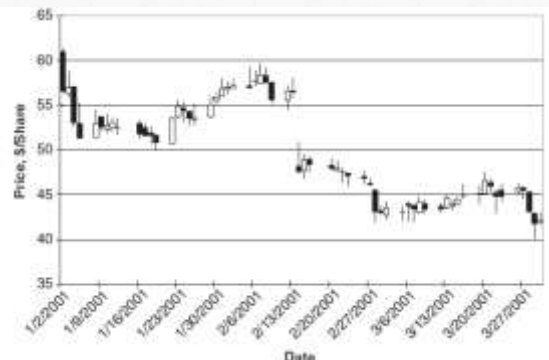


URE 2.3 Scatter plot of temperature anomaly versus CO₂ concentration

GRAPHICAL DISPLAYS - Time Series Plots

A variation of time series plot - Candlestick Chart:

- The following graph displays daily price information for Whole Foods Market stock during the first quarter of 2001 (the trading days from January 2, 2001 through March 30, 2001).
- Shows the opening, closing, highest, and lowest prices experienced within a trading day for the first quarter.
- If the opening price was higher than the closing price, the box is filled;
- If the closing price was higher than the opening price, the box is open.
- More useful than a time series plot of just the closing (or opening) prices, because it shows the volatility of the stock within a trading day.



GRAPHICAL DISPLAYS – Plotting Smoothed Data

- Data smoothing → a statistical technique that removes noise from data to reveal patterns and trends → eliminates outliers and statistical noise.
 - A smoothed version of the original data is overlaid on the original time series plot to help reveal patterns in the original data.
1. A simple data-smoother → simple moving average
 2. An effective data-smoother → Odd-span moving medians (also called running medians)

GRAPHICAL DISPLAYS – Plotting Smoothed Data

Simple moving average

- A simple moving average of span N assigns weights $1/N$ to the most recent N observations $y_T, y_{T-1}, \dots, y_{T-N+1}$, and weight zero to all other observations.
- The N-span moving average at time period T is

$$M_T = \frac{y_T + y_{T-1} + \dots + y_{T-N+1}}{N} = \frac{1}{N} \sum_{t=T-N+1}^T y_t$$

- As each new observation becomes available it is added into the sum from which the moving average is computed and the oldest observation is discarded.
- The moving average reduces variability because it averages out short-term spikes or dips, leaving a smoother representation of the underlying trend in the data

GRAPHICAL DISPLAYS – Plotting Smoothed Data

$$M_T = \frac{y_T + y_{T-1} + \dots + y_{T-N+1}}{N} = \frac{1}{N} \sum_{t=T-N+1}^T y_t$$

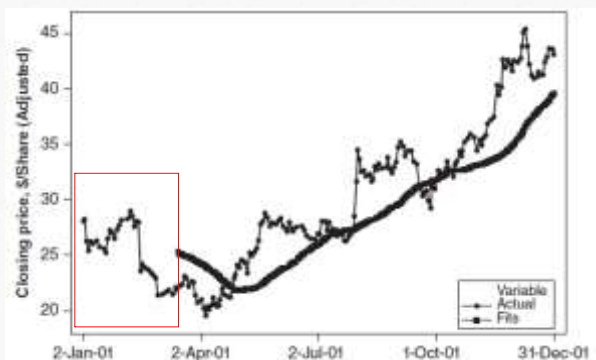
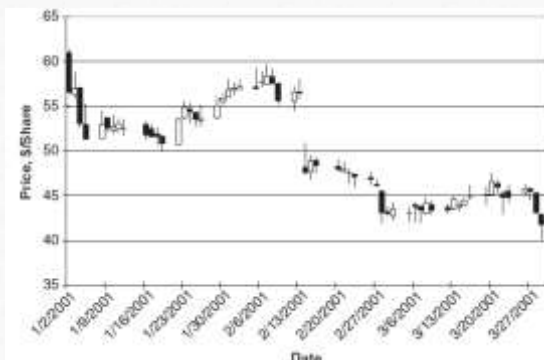
- If the variance of an individual observation y_t $\text{Var}(y_t) = \sigma^2$,

$$\text{Var}(M_T) = \text{Var}\left(\frac{1}{N} \sum_{t=T-N+1}^T y_t\right) = \frac{1}{N^2} \sum_{t=T-N+1}^T \text{Var}(y_t) = \frac{\sigma^2}{N}$$

- Example:
- Original Observations: [10, 12, 18, 15, 20]
- 3-Point Moving Average:
- $MA1=(10+12+18)/3=13.33$ $MA2=(12+18+15)/3=15$ $MA3=(18+15+20)/3=17.67$
- Original Data Variability: Values jump between 10 and 20 (high variability).
- Moving Average Variability: Smoothed values (13.33, 15, 17.67) fluctuate less.

GRAPHICAL DISPLAYS – Plotting Smoothed Data

- Plots of moving averages → used by analysts to evaluate stock price trends → common MA periods are 5, 10, 20, 50, 100, and 200 days.
- A time series plot of Whole Foods Market stock price with a 50-day moving average → smoothens the day-to-day noise and shows a generally increasing trend



GRAPHICAL DISPLAYS – Plotting Smoothed Data

- The simple moving average is a linear data smoother, or a linear filter → because it replaces each observation y_t with a linear combination of the other data points that are near to it in time.
- Hanning filter → applies a weighted averaging process to the data, giving more weight to the central points and gradually reducing the weight as you move towards the edges of the window → used to smooth weather data.

$$M_t^H = 0.25y_{t+1} + 0.5y_t + 0.25y_{t-1}$$

- Disadvantage of a linear filter → an unusual or erroneous data point or an outlier will dominate the moving averages that contain that observation, contaminating the moving averages for a length of time equal to the span of the filter.
- Eg. in the sequence of observations 15, 18, 13, 12, 16, 14, 16, 17, 18, 15, 18, 200, 19, 14, 21, 24, 19, 25, the value 200 should not be analyzed along with the rest of the data.

GRAPHICAL DISPLAYS – Plotting Smoothed Data

Odd-span moving medians (or running medians)

- Effective data smoother when the time series may be contaminated with unusual values or outliers.
- The moving median of span N, where $N = 2u + 1$

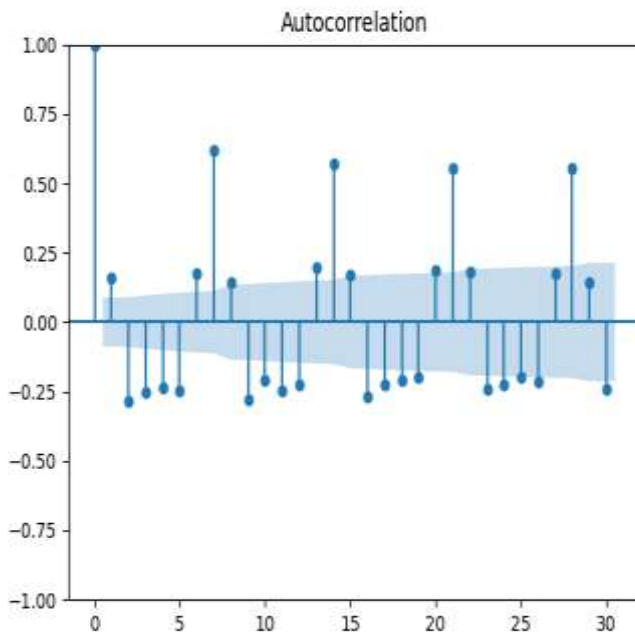
$$m_t^{[N]} = \text{med}(y_{t-u}, \dots, y_t, \dots, y_{t+u})$$

- The moving median of span 3 $m_t^{[3]} = \text{med}(y_{t-1}, y_t, y_{t+1})$
- 3 values processed at a time and 3 original observations replaced by their median.
- Eg. original observation
15, 18, 13, 12, 16, 14, 16, 17, 18, 15, 18, 200, 19, 14, 21, 24, 19, 25
- The smoothed data →
_____, 15, 13, 13, 14, 16, 16, 17, 17, 18, 18, 19, 19, 19, 21, 21, 24, _____
- End values lost.
- Can be applied more than once if desired to obtain an even smoother series of observations
- _____, _____, 13, 13, 14, 16, 16, 17, 17, 18, 18, 19, 19, 19, 21, 21, _____, _____

GRAPHICAL DISPLAYS – Plotting Smoothed Data

Odd-span moving medians (or running medians)

- Repeated application of the moving median will not change the data, apart from the end values.
- The information loss from the missing end values is not serious.
- To maintain the same length, end values copied
- 15, 18, 13, 13, 14, 16, 16, 17, 17, 18, 18, 19, 19, 19, 21, 21, 19, 25
- Smoothing the end values also done using clever methods.



NUMERICAL DESCRIPTION OF TIME SERIES DATA

- Stationary Time Series
- Autocovariance and Autocorrelation Functions
- The Variogram

NUMERICAL DESCRIPTION OF TIME SERIES DATA

- The numerical description of time series data involves summarizing sequential data points using core metrics like mean, variance, autocovariance, and autocorrelation. These numbers measure the central value, spread, and how data points relate to past values over time.
- Mean, variance, standard deviation → statistics-based
- Autocovariance, Autocorrelation (ACF), lag → time series-based
- Three important concepts are revealed using the numerical descriptions
 1. Stationary Time Series
 2. Autocovariance and Autocorrelation Functions
 3. The Variogram

NUMERICAL DESCRIPTION OF TIME SERIES DATA – **Stationary Time Series**

- A time series is said to be strictly stationary if its properties are not affected by a change in the time origin → a time series whose statistical properties do not change over time.
- In a stationary time series

Joint probability distribution
of the observations

$y_t, y_{t+1}, \dots, y_{t+n}$

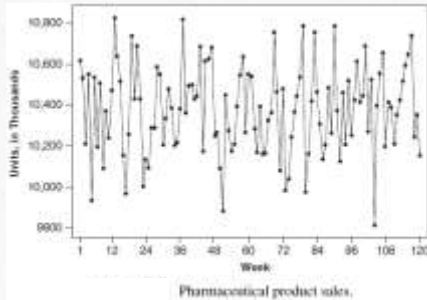
=

Joint probability distribution
of the observations

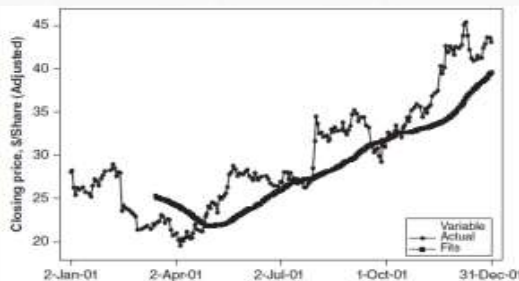
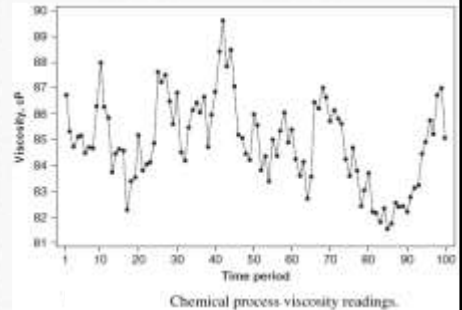
$y_{t+k}, y_{t+k+1}, \dots, y_{t+k+n}$

- Eg.

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Stationary Time Series



Both time series seem to vary around a fixed level → stationary time series



No obvious fixed level → nonstationary time series

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Stationary Time Series

- Stationary implies a type of statistical equilibrium or stability in the data → the time series has a constant mean and a constant variance defined as below:

$$\mu_y = E(y) = \int_{-\infty}^{\infty} yf(y)dy$$

$$\sigma_y^2 = \text{Var}(y) = \int_{-\infty}^{\infty} (y - \mu_y)^2 f(y)dy$$

- $E(y)$ → expectation operator
- $f(y)$ → the probability density function (PDF) of the continuous random variable y .
- Sample mean and sample variance for T observations $y_1, y_2, \dots, y_T$:

$$\bar{y} = \hat{\mu}_y = \frac{1}{T} \sum_{t=1}^T y_t$$

$$s^2 = \hat{\sigma}_y^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2$$

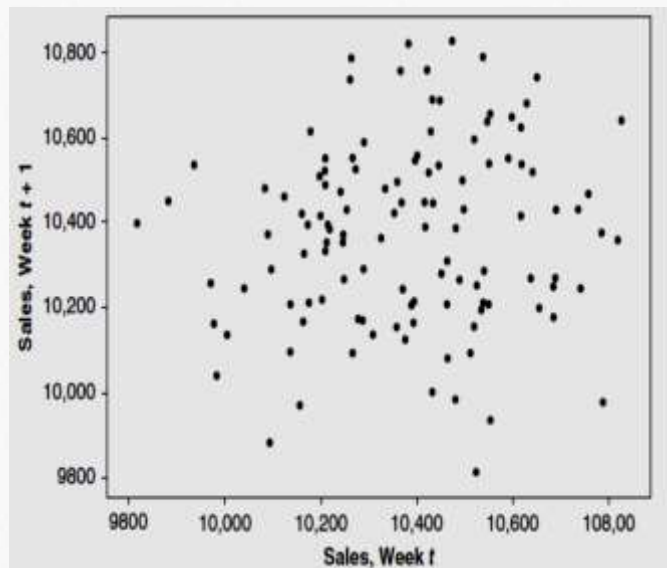
- Considering denominator T → Since T is not usually small, $T-1$ and T are expected to be similar

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

- If a time series is stationary then the joint probability distribution of any two observations, say, y_t and y_{t+k} , is the same for any two time periods t and $t + k$ that are separated by the same interval $k \rightarrow$ The interval k is called the lag.
- Joint probability distribution of any two observations $\rightarrow$ describes the likelihood that both observations take on specific values or fall within specific ranges at the same time.
- Scatter plot of all pairs of (y_t, y_{t+k}) used to represent the nature of the time series.
- Both of the following examples use $k=1$.

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

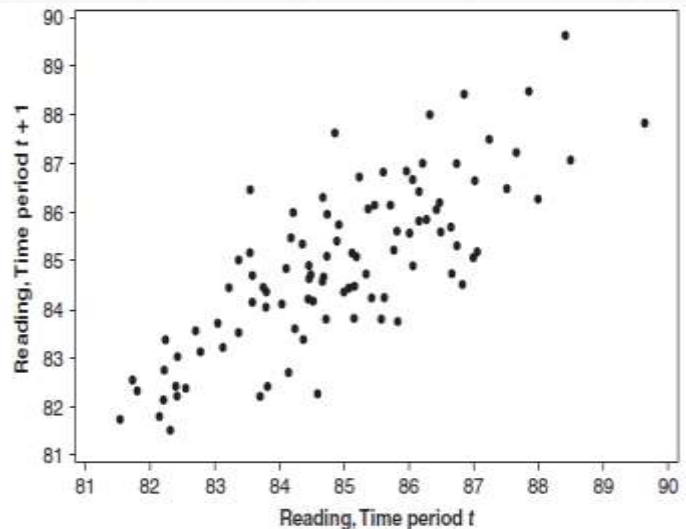
- Uncorrelated adjacent observations $(y_t, y_{t+1}) \rightarrow$ value of y in the current period does not provide any useful information about the value of y that will be observed in the next period



Scatter diagram of pharmaceutical product sales at lag $k = 1$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

- Positively correlated adjacent observations $(y_t, y_{t+1}) \rightarrow$ small value of y_t tends to be followed in the next time period by another small value of y , and a large value of y_t tends to be followed immediately by another large value of y .



Scatter diagram of chemical viscosity readings at lag $k = 1$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

- Autocovariance at lag $k \rightarrow$ The covariance between y_t and its value at another time period, say, $y_{t+k} \rightarrow$ defined by

$$\gamma_k = \text{Cov}(y_t, y_{t+k}) = E[(y_t - \mu)(y_{t+k} - \mu)]$$

- Autocovariance function $\rightarrow$ The collection of the values of γ_k , $k = 0, 1, 2, \dots$
- Autocorrelation coefficient at lag k for a stationary time series $\rightarrow$ measures the correlation between observations separated by lag k

$$\rho_k = \frac{E[(y_t - \mu)(y_{t+k} - \mu)]}{\sqrt{E[(y_t - \mu)^2]E[(y_{t+k} - \mu)^2]}} = \frac{\text{Cov}(y_t, y_{t+k})}{\text{Var}(y_t)} = \frac{\gamma_k}{\gamma_0}$$

- Autocorrelation function (ACF) $\rightarrow$ The collection of the values of ρ_k , $k = 0, 1, 2, \dots \rightarrow$ dimensionless quantity $\rightarrow$ independent of the scale of measurement of the time series

- At lag $k=0 \rightarrow \gamma_0 = \sigma_y^2 \quad \rho_0 = 1 \quad \rho_k = \rho_{-k} \leftarrow \text{ACF is symmetric}$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

Covariance

$$cov_{x,y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

Correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Autocovariance

$$\gamma_k = \text{Cov}(y_t, y_{t+k}) = E[(y_t - \mu)(y_{t+k} - \mu)]$$

Auto Correlation

$$\rho_k = \frac{E[(y_t - \mu)(y_{t+k} - \mu)]}{\sqrt{E[(y_t - \mu)^2]E[(y_{t+k} - \mu)^2]}} = \frac{\text{Cov}(y_t, y_{t+k})}{\text{Var}(y_t)} = \frac{\gamma_k}{\gamma_0}$$

Estimate the autocovariance

$$c_k = \hat{\gamma}_k = \frac{1}{T} \sum_{t=1}^{T-k} (y_t - \bar{y})(y_{t+k} - \bar{y}), \quad k = 0, 1, 2, \dots, K$$

Sample autocorrelation function

$$r_k = \hat{\rho}_k = \frac{c_k}{c_0}, \quad k = 0, 1, \dots, K$$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

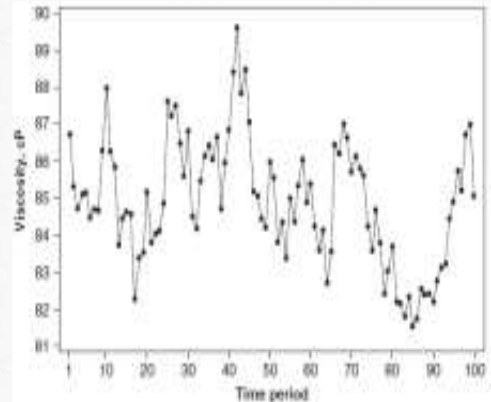
- Necessary to estimate the autocovariance and ACFs from a time series of finite length, say, $y_1, y_2, \dots, y_T$
- At least 50 observations are required to give a reliable estimate of the ACF,
- The individual sample autocorrelations should be calculated up to lag K , where K is about $T/4$.
- To identify the k , for which $\rho_k=0$, the r_k is compared to its standard error $\rightarrow se(r_k)$

$$se(r_k) \cong \frac{1}{\sqrt{T}}$$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

TABLE 2.1 Chemical Process Viscosity Readings

Time Period	Reading	Time Period	Reading	Time Period	Reading	Time Period	Reading
1	86.7418	26	87.2397	51	85.5722	76	84.7052
2	83.3195	27	87.5219	52	83.7935	77	83.8168
3	84.7355	28	86.4992	53	84.3706	78	82.4171
4	85.1113	29	85.6050	54	83.3762	79	83.0420
5	85.1487	30	86.8293	55	84.9975	80	83.6993
6	84.4775	31	84.5004	56	84.3495	81	82.2033
7	84.6827	32	84.1844	57	85.3395	82	82.1413
8	84.6757	33	85.4563	58	86.0503	83	81.7961
9	86.3169	34	86.1511	59	84.8839	84	82.3241
10	88.0006	35	86.4142	60	85.4176	85	81.5316
11	86.2597	36	86.0498	61	84.2309	86	81.7280
12	85.8286	37	86.6642	62	83.5761	87	82.5375
13	83.7500	38	84.7289	63	84.1343	88	82.3877
14	84.4628	39	85.9523	64	82.6974	89	82.4159
15	84.6476	40	86.8473	65	83.5454	90	82.2102
16	84.5751	41	88.4250	66	86.4714	91	82.7673
17	82.2473	42	89.6481	67	86.2143	92	83.1234
18	83.3774	43	87.8566	68	87.0215	93	83.2203
19	83.5385	44	88.4997	69	86.6504	94	84.4510
20	85.1620	45	87.0622	70	85.7082	95	84.9145
21	83.7881	46	85.1973	71	86.1504	96	85.7609
22	84.0421	47	85.0767	72	85.8032	97	85.2302
23	84.1023	48	84.4362	73	85.6197	98	86.7312
24	84.8495	49	84.2112	74	84.2339	99	87.0048
25	87.6416	50	85.9952	75	83.5737	100	85.0572



Chemical process viscosity readings.

Eg. problem

- Consider the chemical process viscosity readings plotted in Figure; the values are listed in Table 2.1. Calculate the sample ACF at lag $k = 1$.

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

$$c_k = \hat{\gamma}_k = \frac{1}{T} \sum_{t=1}^{T-k} (y_t - \bar{y})(y_{t+k} - \bar{y}), \quad k = 0, 1, 2, \dots, K$$

$$r_k = \hat{\rho}_k = \frac{c_k}{c_0}, \quad k = 0, 1, \dots, K$$

- Given lag $k=1$.
- So calculate c_k for $k=0, k=1$.
- Then calculate r_k .

$$\begin{aligned} c_0 &= \frac{1}{100} \sum_{t=1}^{100-0} (y_t - \bar{y})(y_{t+0} - \bar{y}) \\ &= \frac{1}{100} [(86.7418 - 84.9153)(86.7418 - 84.9153) + \dots \\ &\quad + (85.0572 - 84.9153)(85.0572 - 84.9153)] \\ &= 280.9332 \end{aligned}$$

$$\begin{aligned} c_1 &= \frac{1}{100} \sum_{t=1}^{100-1} (y_t - \bar{y})(y_{t+1} - \bar{y}) \\ &= \frac{1}{100} [(86.7418 - 84.9153)(85.3195 - 84.9153) + \dots \\ &\quad + (87.0048 - 84.9153)(85.0572 - 84.9153)] \\ &= 220.3137 \end{aligned}$$

$$r_1 = \frac{c_1}{c_0} = \frac{220.3137}{280.9332} = 0.7842$$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

- Behaviour of stationary time series:
- The rate of decrease or decay in ACF values in Figure 1 from 0.78 to 0, followed by a sinusoidal pattern about 0.
- The sample ACFs for pharmaceutical product sales plotted in Figure 2 appear randomly positive or negative, with values near zero.

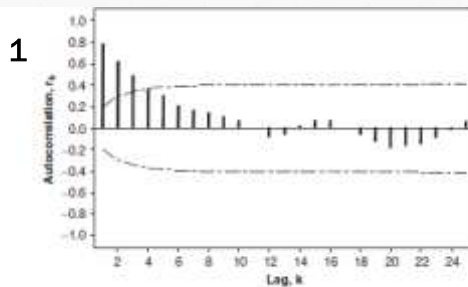


FIGURE 2.12 Sample autocorrelation function for chemical viscosity readings, with 5% significance limits.

Autocorrelation function: reading

Lag	ACF	T	LSG
1	0.784251	1.84	83.38
2	0.528850	4.21	134.41
3	0.401867	2.83	120.85
4	0.300003	1.84	143.02
5	0.304254	1.87	153.77
6	0.308879	1.05	157.72
7	0.164820	0.82	181.48
8	0.144789	0.72	183.88
9	0.103625	0.81	185.01
10	0.060209	0.21	193.51
11	0.003948	0.02	195.01
12	0.077225	0.33	195.23
13	-0.057563	-0.85	195.72
14	0.022545	0.10	198.57
15	0.072784	0.38	197.21
16	0.075128	0.33	197.81
17	0.003824	0.05	197.81
18	-0.053525	-0.29	198.22
19	-0.021327	-0.02	199.12
20	-0.185448	-0.88	175.88
21	-0.183488	-0.73	177.71
22	-0.149719	-0.70	180.48
23	-0.083491	-0.43	181.22
24	-0.018598	-0.09	181.51
25	0.003179	0.03	182.26

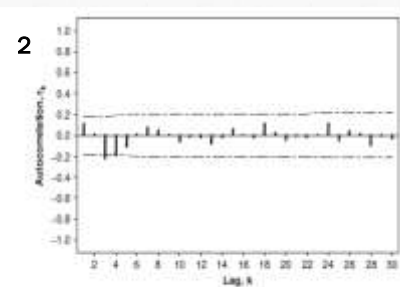


FIGURE 2.14 Autocorrelation function for pharmaceutical product sales, with 5% significance limits.

NUMERICAL DESCRIPTION OF TIME SERIES DATA – Autocovariance and Autocorrelation Functions

Characteristic behaviour of a nonstationary time series

- Instead of cutting off or tailing off near zero after a few lags, this sample ACF is very persistent; that is, it decays very slowly and exhibits sample autocorrelations that are still rather large even at long lags.

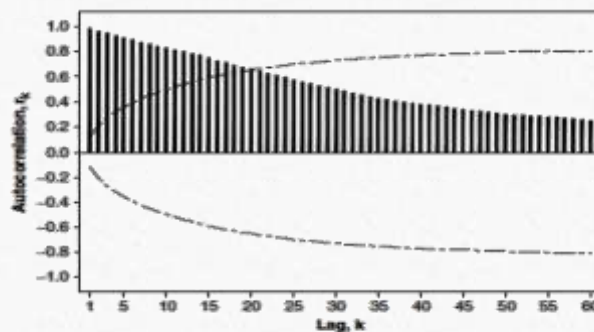


FIGURE 2.15 Autocorrelation function for Whole Foods Market stock price with 5% significance limits.

NUMERICAL DESCRIPTION OF TIME SERIES DATA – The Variogram

- Two techniques for determining if a time series is nonstationary,
 - *Plotting a reasonable long series of the data to see if it drifts or wanders away from its mean for long periods of time,*
 - *Computing the sample ACF.*
- However differentiating stationary & non-stationary processes in real-world time series is not so clear.
- An useful diagnostic tool → the variogram.
- The variogram G_k measures variances of the differences between observations that are k lags apart ($k=0,1,2,\dots$), relative to the variance of the differences that are at lag 1 ($k=1$). Mathematically,

$$G_k = \frac{\text{Var}(y_{t+k} - y_t)}{\text{Var}(y_{t+1} - y_t)} \quad k = 1, 2, \dots$$

Eg. For $t=5$ $\frac{\text{Var}(y_6 - y_5)}{\text{Var}(y_6 - y_5)}, \frac{\text{Var}(y_7 - y_5)}{\text{Var}(y_6 - y_5)}, \frac{\text{Var}(y_8 - y_5)}{\text{Var}(y_6 - y_5)}$

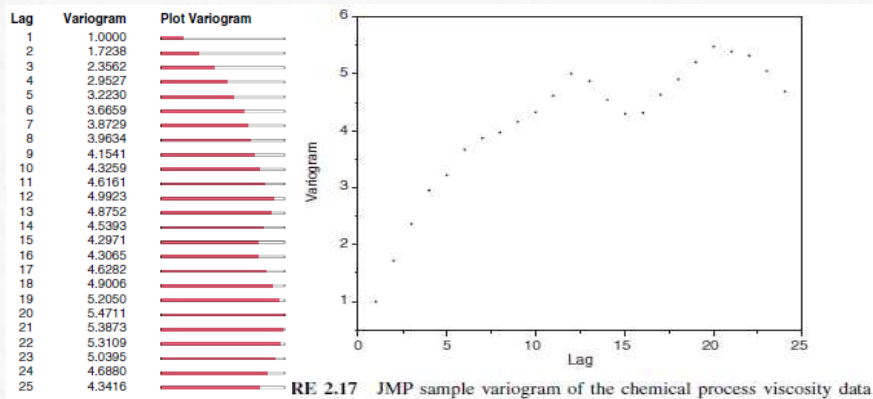
NUMERICAL DESCRIPTION OF TIME SERIES DATA – The Variogram

- Values of G_k are plotted as a function of the lag k .
- If the time series is stationary, then $G_k = \frac{1 - \rho_k}{1 - \rho_1}$
- Autocorrelation $\rho_k \rightarrow$ measures the correlation between observations separated by lag $k \rightarrow$ measures how past values influence future values at different lags k .
- As k increases, the correlation between observations that are far apart weakens, eventually approaching zero. $G_k = 1/(1 - \rho_1)$
- So for a stationary time series $\rho_k = 0$, as k increases. Hence
- Using an estimate of $\text{Var}(y_{t+k} - y_t)$ as s_k^2 , The sample variogram is given by

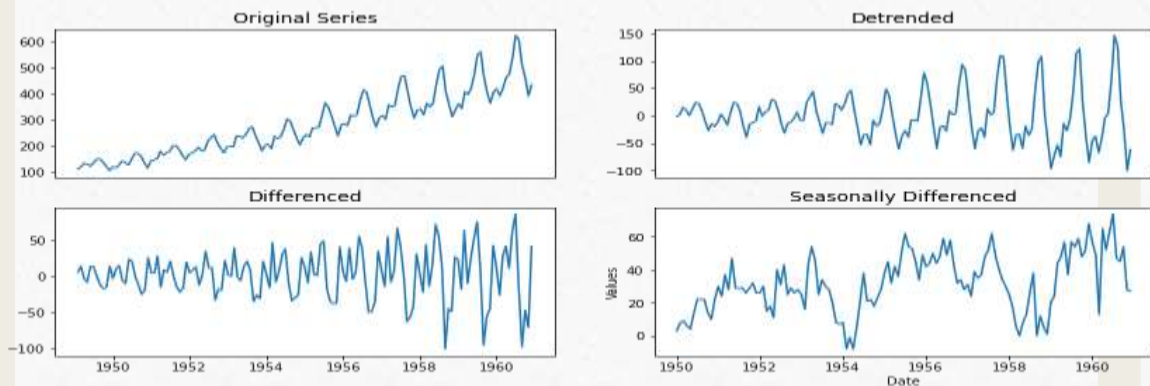
$$\hat{G}_k = \frac{s_k^2}{s_1^2} \quad k = 1, 2, \dots$$

$$d_t^k = y_{t+k} - y_t \quad \bar{d}^k = \frac{1}{T-k} \sum d_t^k \quad s_k^2 = \frac{\sum_{t=1}^{T-k} (d_t^k - \bar{d}^k)^2}{T-k-1}$$

NUMERICAL DESCRIPTION OF TIME SERIES DATA – The Variogram



- The sample variogram generally converges to a stable level and then fluctuates around it.



USE OF DATA TRANSFORMATIONS AND ADJUSTMENTS

- Transformations
- Trend and Seasonal Adjustments

Data Transformations

- Data transformations → the process of applying mathematical operations to the data to make it more suitable for modeling and forecasting → useful in statistical work for stabilizing the variance of the data.
- Goal → to stabilize variance, remove trends, make data stationary, and improve interpretability.
- A popular type of data transformation to deal with non-constant variance → the power family of transformations.

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda y^{\lambda-1}}, & \lambda \neq 0 \\ y \ln y, & \lambda = 0 \end{cases}, \quad \hat{y} = \exp[(1/T) \sum_{t=1}^T \ln y_t]$$

- Typical values of λ used with time series data
 - $\lambda = 0.5$ (a square root transformation),
 - $\lambda = 0$ (the log transformation),
 - $\lambda = -0.5$ (reciprocal square root transformation), and
 - $\lambda = -1$ (inverse transformation).
- $\lambda=1 \rightarrow$ no transformation.

Data Transformations

- How to choose an appropriate value of λ ?
 - *By fitting a model to $y^{(\lambda)}$*
 - *Then selecting the transformation that produces the minimum residual sum of squares $\sum_{t=1}^n (y_t - \hat{y}_t)^2$. Choose λ that causes the minimum error*

- The approximate percentage change in y_t can be calculated from the differences of the log-transformed time series

$$\begin{aligned} 100[\ln(y_t) - \ln(y_{t-1})] &= 100 \ln \left(\frac{y_t}{y_{t-1}} \right) = 100 \ln \left(\frac{y_{t-1} + (y_t - y_{t-1})}{y_{t-1}} \right) \\ &= 100 \ln \left(1 + \frac{x_t}{100} \right) \cong x_t \end{aligned}$$

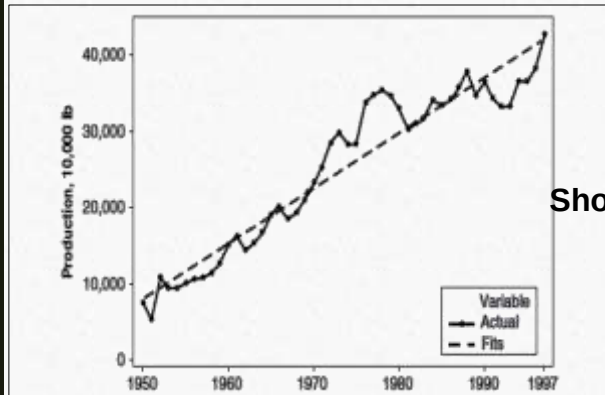
Trend and Seasonal Adjustments

- Used in addition to data transformations.
- Widely used adjustments → trend adjustments and seasonal adjustments (or) trend and seasonal decomposition.
- A time series that exhibits a trend is a nonstationary time series.
- Modeling and forecasting of such a time series is greatly simplified if the trend can be eliminated → by fitting a regression model → and then subtracting trend component to the data out of the original observations → leaves a set of residuals that are free of trend.
- Approaches: 1. fit a regression model, 2. by differencing the data
- 1. Linear trend: $E(y_t) = \beta_0 + \beta_1 t$ $E(y_t) = \beta_0 + \beta_1 t + \beta_2 t^2$ $E(y_t) = \beta_0 e^{\beta_1 t}$
- Eg. Blue and gorgonzola cheese production

Trend and Seasonal Adjustments

1. Linear trend: $E(y_t) = \beta_0 + \beta_1 t$ $E(y_t) = \beta_0 + \beta_1 t + \beta_2 t^2$ $E(y_t) = \beta_0 e^{\beta_1 t}$
- In **time series analysis**, a **linear trend** refers to a consistent upward or downward movement in the data over time → can be modeled with a straight line → this trend shows that the variable of interest increases or decreases **at a constant rate** over time.
 - A linear trend is typically expressed as: $Y_t = a + bt$ where
 - Y_t = value of the time series at time t
 - a = intercept (value when $t=0$)
 - t = time period
 - b = slope (rate of increase or decrease per time unit)
 - ❖ If $b > 0$: **Upward trend** (values increase over time)
 - ❖ If $b < 0$: **Downward trend** (values decrease over time)
 - ❖ If $b = 0$: **No trend** (values stay constant on average)
 - Eg. Blue and gorgonzola cheese production
 - Linear trends help in:
 - Understanding long-term movements
 - Forecasting future values
 - Decomposing time series into trend, seasonal, and irregular components

Trend and Seasonal Adjustments

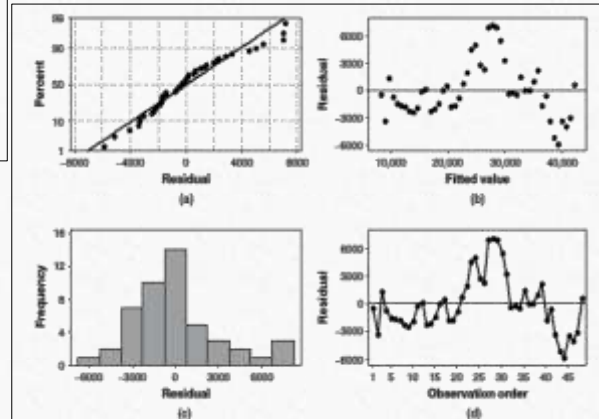


A line fitted through the data points in a **line plot** can help visually identify whether a linear trend exists.

Shows a positive, nearly linear trend

The normal probability plot (a) and histogram (c) indicate the residuals are approximately normally distributed

The plots of residuals versus fitted values (b) and versus observation order (d) indicate non-constant variance in the last half of the time series



Trend and Seasonal Adjustments

- 2. By differencing the data → in 2 ways
 - i. By applying the difference operator to the original time series to obtain a new time series, where ∇ is the (backward) difference operator. $x_t = y_t - y_{t-1} = \nabla y_t$
 - ii. In terms of a backshift operator B , where $By_t = y_{t-1}$
- Hence $x_t = y_t - By_t$ $x_t = (1 - B)y_t = \nabla y_t = y_t - y_{t-1}$ So $\nabla = (1 - B)$
- Successive differencing done if necessary until the trend is removed.

$$x_t = \nabla^2 y_t = \nabla(\nabla y_t) = (1 - B)^2 y_t = (1 - 2B + B^2)y_t = y_t - 2y_{t-1} + y_{t-2}$$

- In general

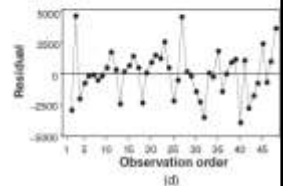
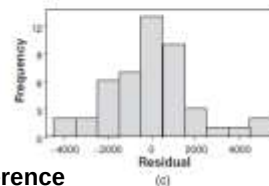
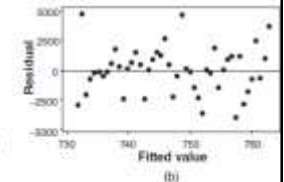
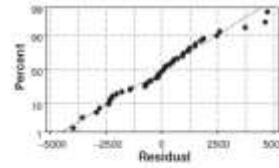
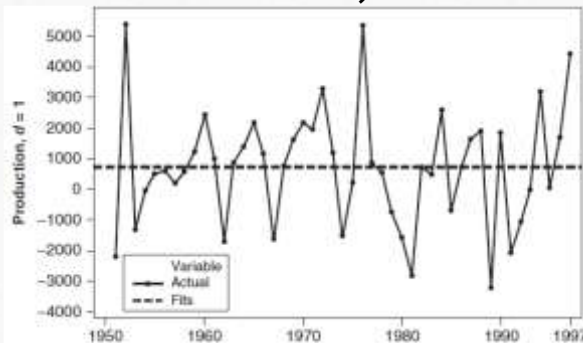
$$B^d y_t = y_{t-d}$$

$$\nabla^d = (1 - B)^d$$

Trend and Seasonal Adjustments

■ Advantages of differencing:

- *it does not require estimation of any parameters, so it is a more parsimonious (i.e., simpler) approach →*
- *model fitting assumes that the trend is fixed (or deterministic) throughout the time series history and will remain so in the (at least immediate) future.*



Blue and gorgonzola cheese production, with one difference

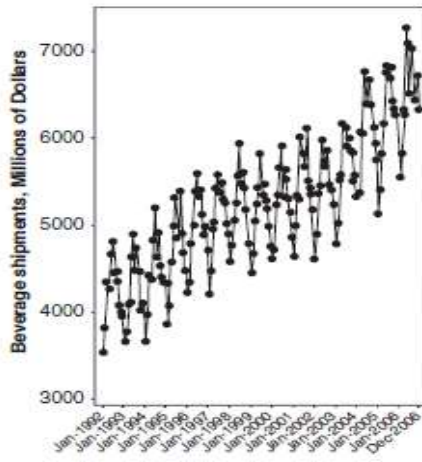
Trend and Seasonal Adjustments

- Differencing can also be used to eliminate seasonality.
- A lag-d seasonal difference operator can be defined as

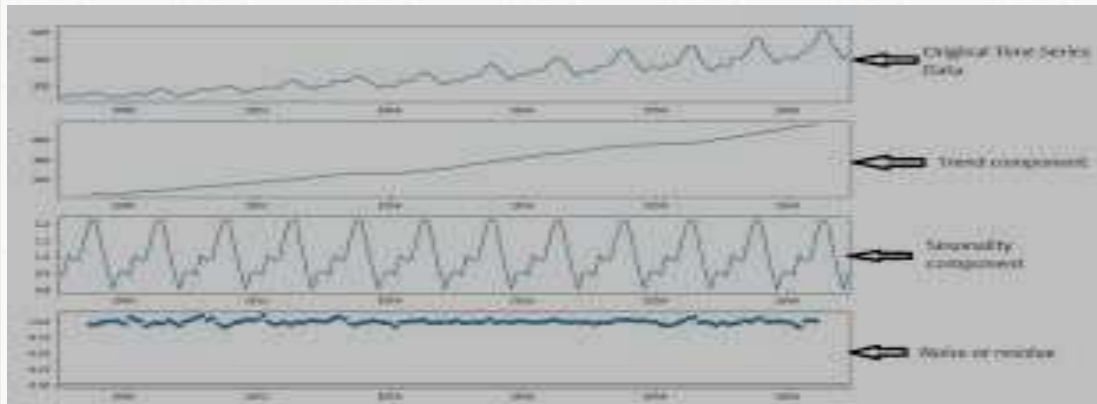
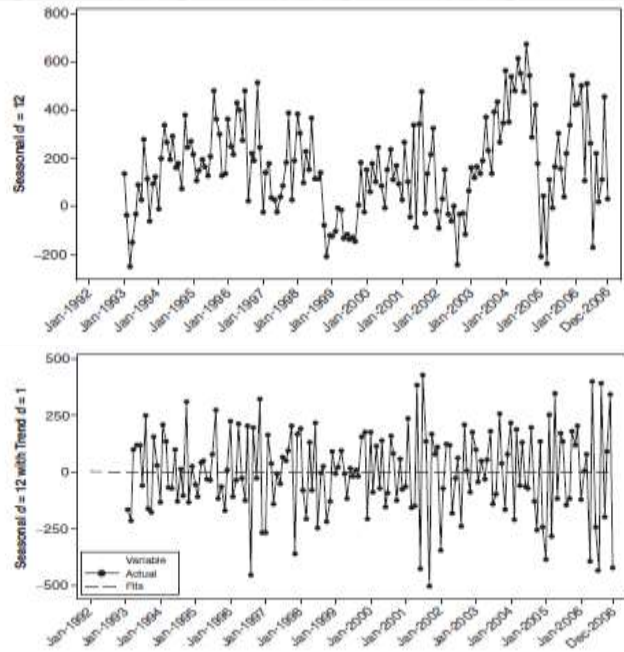
$$\nabla_d y_t = (1 - B^d) y_t = y_t - y_{t-d}$$

- When both trend and seasonal components are simultaneously present, it can be sequentially differenced to eliminate these effects.
 - first seasonally difference to remove the seasonal component
 - then difference one or more times using the regular difference operator to remove the trend.

Trend and Seasonal Adjustments



Time series plots of seasonal- and trend-differenced beverage data.



GENERAL APPROACH TO TIME SERIES MODELLING AND FORECASTING

GENERAL APPROACH TO TIME SERIES MODELLING AND FORECASTING

- The basic steps in modeling and forecasting a time series:
 1. Plot the time series and determine its basic features, such as whether trends or seasonal behavior or both are present. Look for possible outliers or any indication that the time series has changed with respect to its basic features (such as trends or seasonality) over the time period history.
 2. Eliminate any trend or seasonal components, either by differencing or by fitting an appropriate model to the data. Also consider using data transformations, particularly if the variability in the time series seems to be proportional to the average level of the series. The objective of these operations is to produce a set of stationary residuals.

GENERAL APPROACH TO TIME SERIES MODELLING AND FORECASTING

3. Develop a forecasting model for the residuals. It is not unusual to find that there are several plausible models, and additional analysis will have to be performed to determine the best one to deploy. Sometimes potential models can be eliminated on the basis of their fit to the historical data. It is unlikely that a model that fits poorly will produce good forecasts.
4. Validate the performance of the model (or models) from the previous step. This will probably involve some type of split-sample or cross validation procedure. The objective of this step is to select a model to use in forecasting.

GENERAL APPROACH TO TIME SERIES MODELLING AND FORECASTING

5. Also the differences between the original time series y_t and the values that would be forecast by the model on the original scale are also necessary. To forecast values on the scale of the original time series y_t , reverse the transformations and any differencing adjustments made to remove trends or seasonal effects.
6. For forecasts of future values in period $T + \tau$ on the original scale, if a transformation was used, say, $x_t = \ln y_t$, then the forecast made at the end of period T for $T + \tau$ would be obtained by reversing the transformation. For the natural log this would be

$$\hat{y}_{T+\tau}(T) = \exp[\hat{x}_{T+\tau}(T)]$$

GENERAL APPROACH TO TIME SERIES MODELLING AND FORECASTING

7. If prediction intervals are desired for the forecast (recommended to be done), construct prediction intervals for the residuals and then reverse the transformations made to produce the residuals as described earlier.
8. Develop and implement a procedure for monitoring the forecast to ensure that deterioration in performance will be detected reasonably quickly. Forecast monitoring is usually done by evaluating the stream of forecast errors that are experienced.

EVALUATING AND MONITORING FORECASTING MODEL PERFORMANCE

- **Forecasting Model Evaluation**
- **Choosing Between Competing Models**
- **Monitoring a Forecasting Model**

Forecasting Model Evaluation

- Purpose → To determine how accurately a forecasting model predicts future observations and to continuously check whether its performance remains satisfactory.
- Meaning of performance?
- Using statistical measures → goodness-of-fit approach.
- Uses the residuals and does not really reflect the capability of the forecasting technique to successfully predict future observations.
- The user of the forecasts concerned about the accuracy of future forecasts (forecast accuracy → out-of-sample forecast error), not model goodness of fit.
- So it is important to evaluate this aspect of any recommended technique.
- Model-fitting residuals (error) must be differentiated from out-of-sample forecast error.
- Good fit to historical data does not necessarily mean good forecasting performance.
- Measure of forecast accuracy → always be evaluated as part of a model validation effort and to discriminate between competing models.
- Forecast accuracy should be evaluated using data not used for model fitting.

Forecasting Model Evaluation

- Forecast errors $e_t(1) = y_t - \hat{y}_t(t-1)$ Forecast of y_t
- At several lags can also be considered
- For n observations, forecasts are made and n one-step-ahead forecast errors $e_t(1)$, $t=1, 2, \dots, n$ obtained.
- Standard measures of forecast accuracy are:
 1. Average error or mean error
 2. Mean absolute deviation (or mean absolute error):
 3. Mean squared error

$$ME = \frac{1}{n} \sum_{t=1}^n e_t(1)$$

$$MAD = \frac{1}{n} \sum_{t=1}^n |e_t(1)|$$

$$MSE = \frac{1}{n} \sum_{t=1}^n [e_t(1)]^2$$

Forecasting Model Evaluation

- ME $\rightarrow$ estimate of the expected error value $\rightarrow$ should be close to 0
 $\rightarrow$ indicates biased forecasts are being generated.

$$ME = \frac{1}{n} \sum_{t=1}^n e_t(1)$$

- MAD/MAE $\rightarrow$ SD and variance of the one-step-ahead forecast errors
 $\rightarrow$ measures the variability in forecast errors $\rightarrow$ need to be smaller (better forecast)

$$MAD = \frac{1}{n} \sum_{t=1}^n |e_t(1)|$$

- MSE $\rightarrow$ Gives greater weight to large errors.

$$MSE = \frac{1}{n} \sum_{t=1}^n [e_t(1)]^2$$

Forecasting Model Evaluation

- Percentage-Based Errors →
 1. Mean Percentage Error (MPE) and
 2. Mean Absolute Percentage Error (MAPE)
- MPE and MAPE express forecast accuracy as a percentage and is easier to interpret across different scales.
- Not suitable when actual values contain zero.

$$MPE = \frac{1}{n} \sum \left(\frac{e_t}{y_t} \right) 100$$

$$MAPE = \frac{1}{n} \sum \left| \frac{e_t}{y_t} \right| 100$$

Forecasting Model Evaluation

Ideal Forecast Errors

- Ideally, forecast errors should:
 - *Have mean close to zero (no overpredicting or underpredicting).*
 - *Have small variability*
 - *Show no systematic pattern*
 - *Be approximately white noise*
- The ACF of forecast errors should not show significant spikes.
- If patterns remain in the errors, the forecasting model may be improved.
- A good forecasting model should leave behind errors that are approximately random (white noise), with mean near zero, small variability, no systematic pattern, and no significant autocorrelation.

Choosing Between Competing Models

- Often, several forecasting models can be used for the same time series.
- Do not select a model only because it fits historical data best.
- A model with too many parameters may overfit historical data and perform poorly on new data.
- Data Splitting (or cross-validation) → The data can be divided into:
 - *Training data* → *Model fitting*
 - *Test data* → *Forecast evaluation (at least 20 or 25 observations)*
- The model with the smallest out-of-sample forecast error is generally preferred.

Choosing Between Competing Models

Model Selection Criteria : **Smaller value → better model**

- Common criteria include:
 - MSE
 - Adjusted R^2
 - AIC (Akaike Information Criterion)
 - BIC/SIC (Schwarz Bayesian Information Criterion)
 - AICc – Corrected AIC
- Data splitting/testing should be preferred whenever possible, because it directly evaluates the model's predictive performance on unseen data.
- Principle of parsimony → time-honored model-building principle → all other things being equal, simple models are preferred to complex ones

Monitoring a Forecasting Model

- Even a good forecasting model may become less accurate over time because:
 - The underlying time-series pattern may change.
 - The level or trend may change.
 - Variability may increase.
 - External factors may affect the process.
 - Therefore, the model should be continuously monitored.
- Forecast Errors for Monitoring
- One-step-ahead forecast errors are commonly monitored.
 - If the errors start showing a systematic positive or negative pattern, it may indicate that the model has become biased or inadequate.

Monitoring a Forecasting Model

Shewhart Control Chart

- A simple way to monitor forecast errors.
- It contains:
 - Center line → usually zero or mean forecast error
 - Upper Control Limit (UCL)
 - Lower Control Limit (LCL).

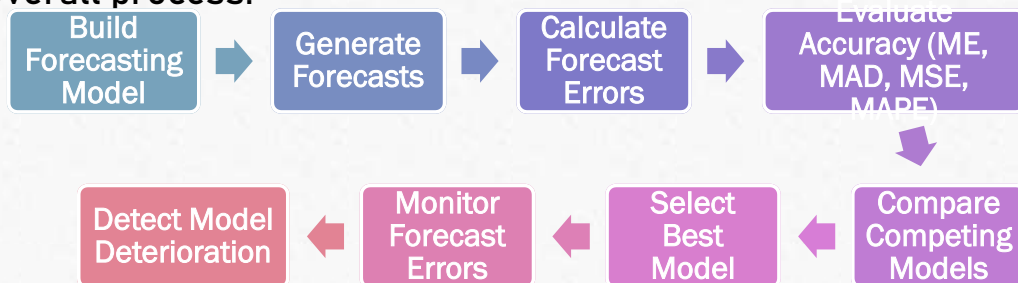


Monitoring a Forecasting Model

Interpretation

- Errors within control limits → Model performance is considered stable.
- Errors outside control limits → Possible model deterioration, unusual observation, or outlier.
- Further investigation is required.

Overall process:



Thank You.

Sales Pitch

2022
Presentation